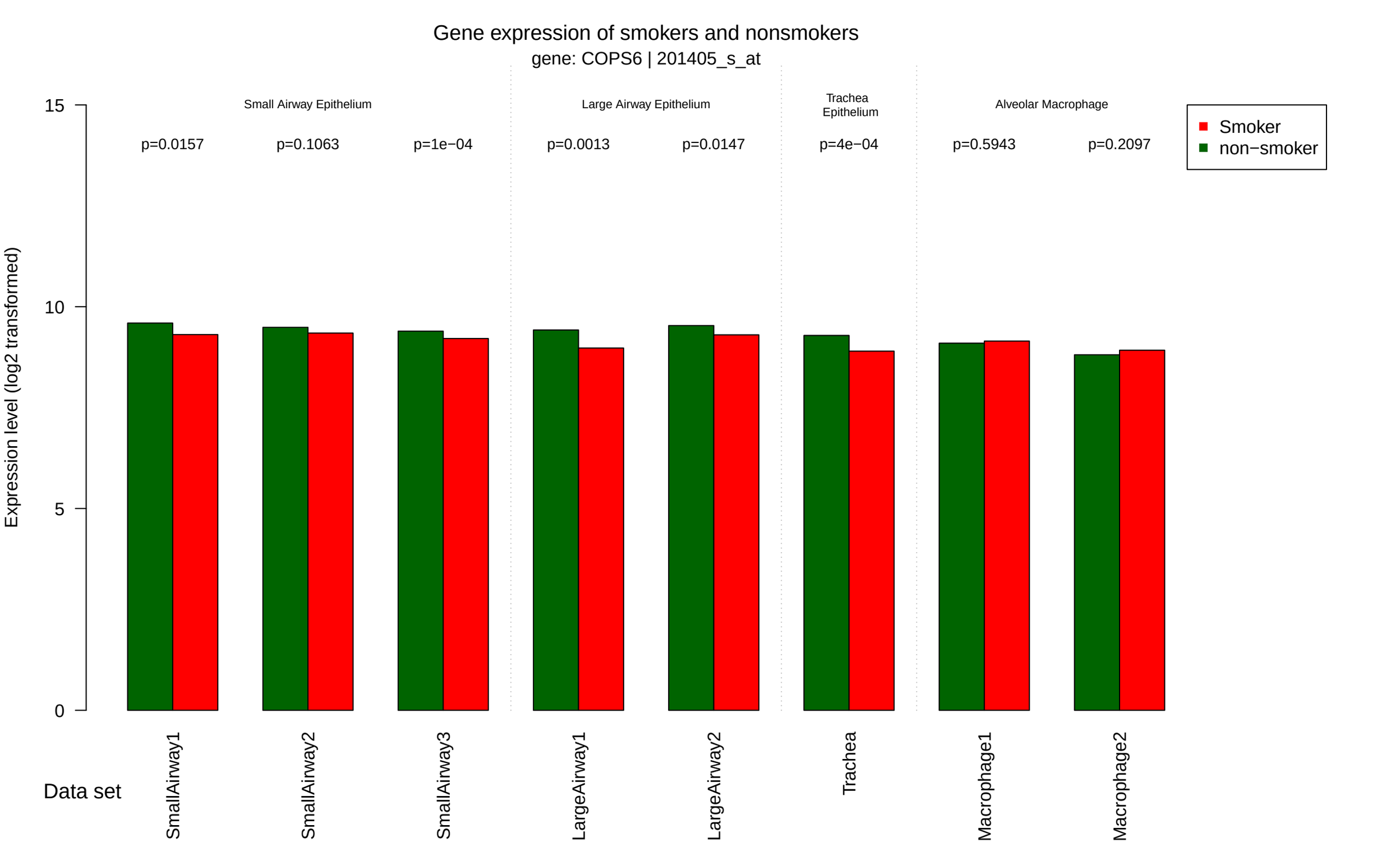
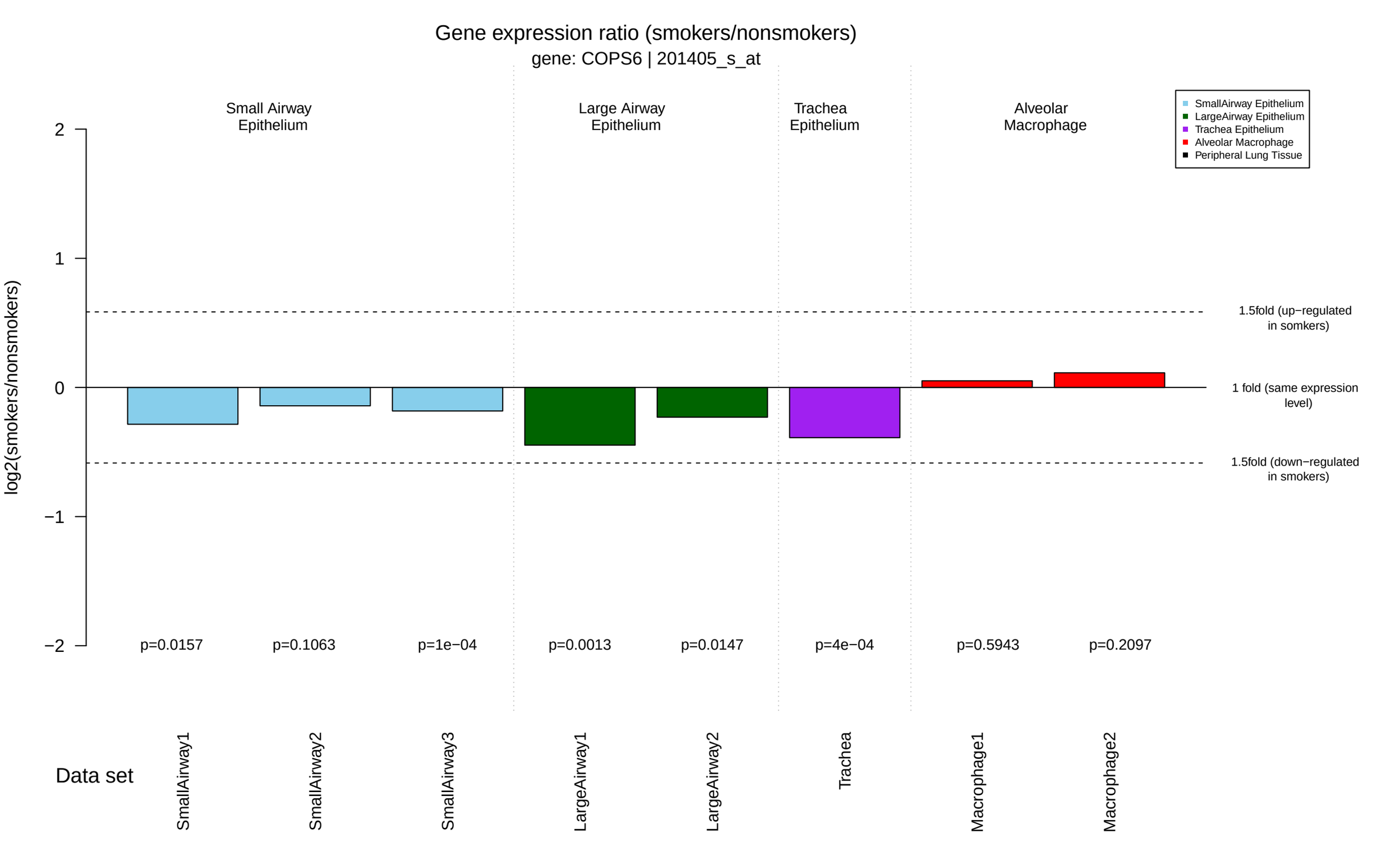
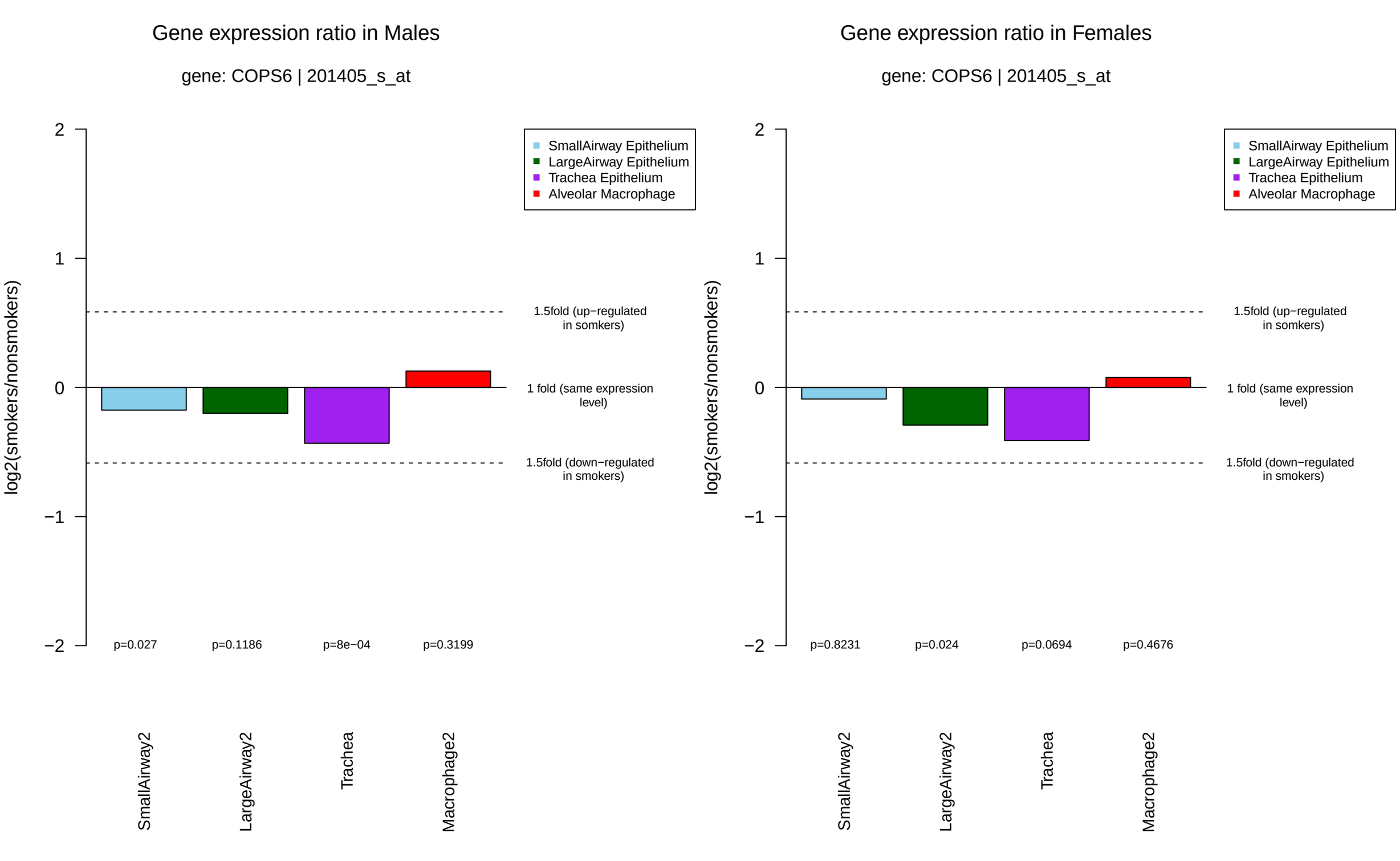
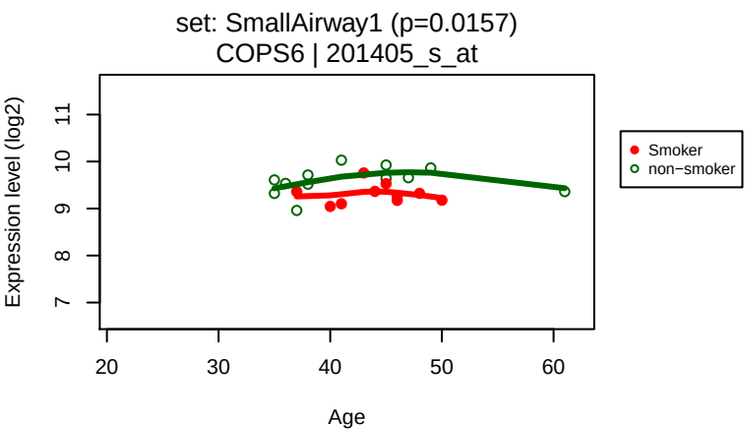
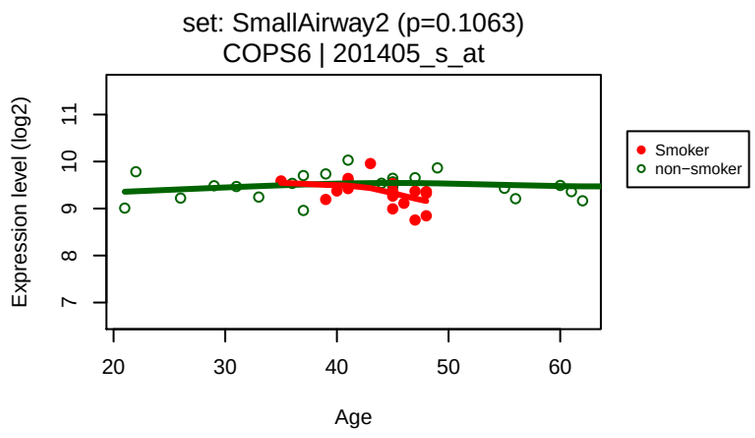
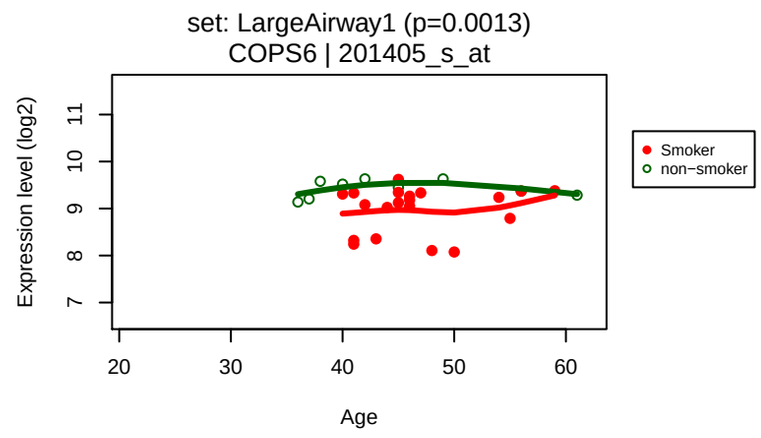

Gene expression of smokers and nonsmokers
gene: COPS6 | 201405_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0157
p=0.1063
p=1e−04
p=0.0013
p=0.0147
p=4e−04
p=0.5943
p=0.2097
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

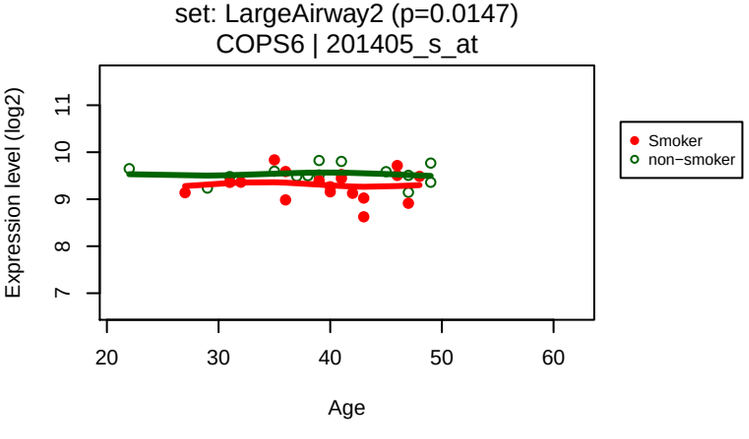

Gene expression ratio (smokers/nonsmokers)
gene: COPS6 | 201405_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0157
p=0.1063
p=1e−04
p=0.0013
p=0.0147
p=4e−04
p=0.5943
p=0.2097
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

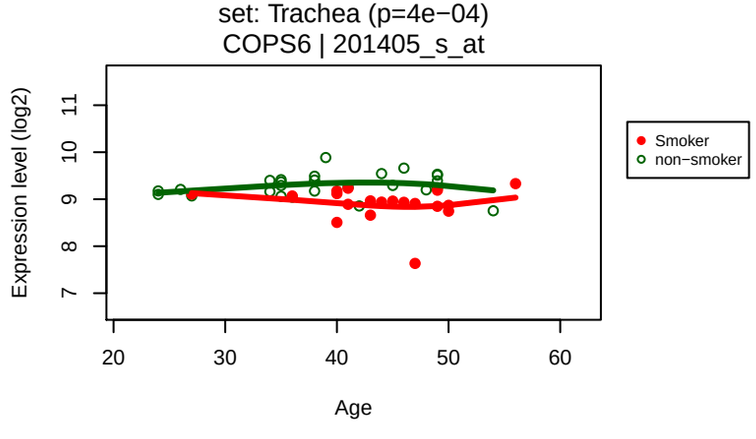

Gene expression ratio in Males
gene: COPS6 | 201405_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.027
p=0.1186
p=8e−04
p=0.3199
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COPS6 | 201405_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8231
p=0.024
p=0.0694
p=0.4676
SmallAirway2
LargeAirway2
Trachea
Macrophage2

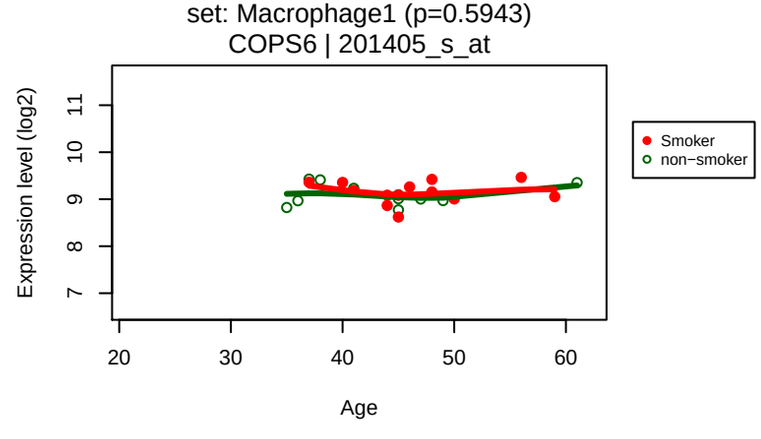

set: SmallAirway1 (p=0.0157)
COPS6 | 201405_s_at
Expression level (log2)
Age
Smoker
non−smoker

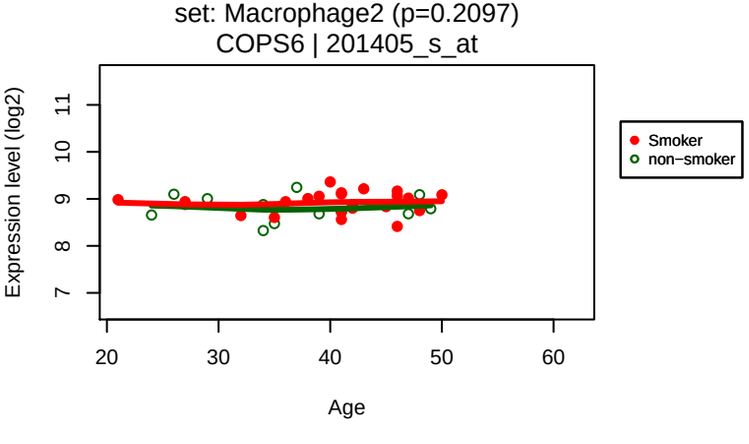

set: SmallAirway2 (p=0.1063)
COPS6 | 201405_s_at
Expression level (log2)
Age
Smoker
non−smoker

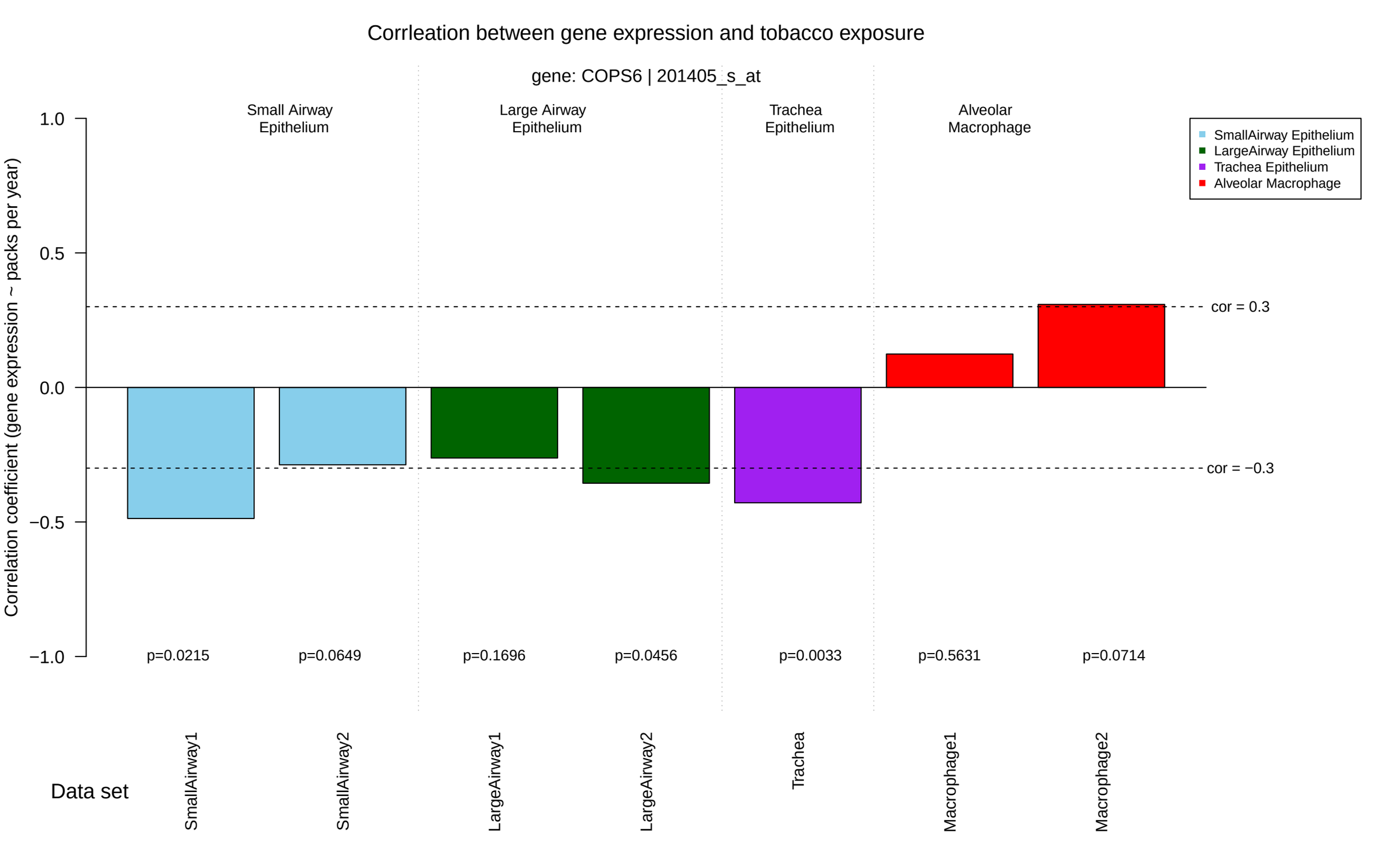

set: LargeAirway1 (p=0.0013)
COPS6 | 201405_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0147)
COPS6 | 201405_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=4e−04)
COPS6 | 201405_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5943)
COPS6 | 201405_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2097)
COPS6 | 201405_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COPS6 | 201405_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0215
p=0.0649
p=0.1696
p=0.0456
p=0.0033
p=0.5631
p=0.0714
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2